

VISAR: Visualization and Interpretation of Sensor-based Activity Recognition for Smart Homes

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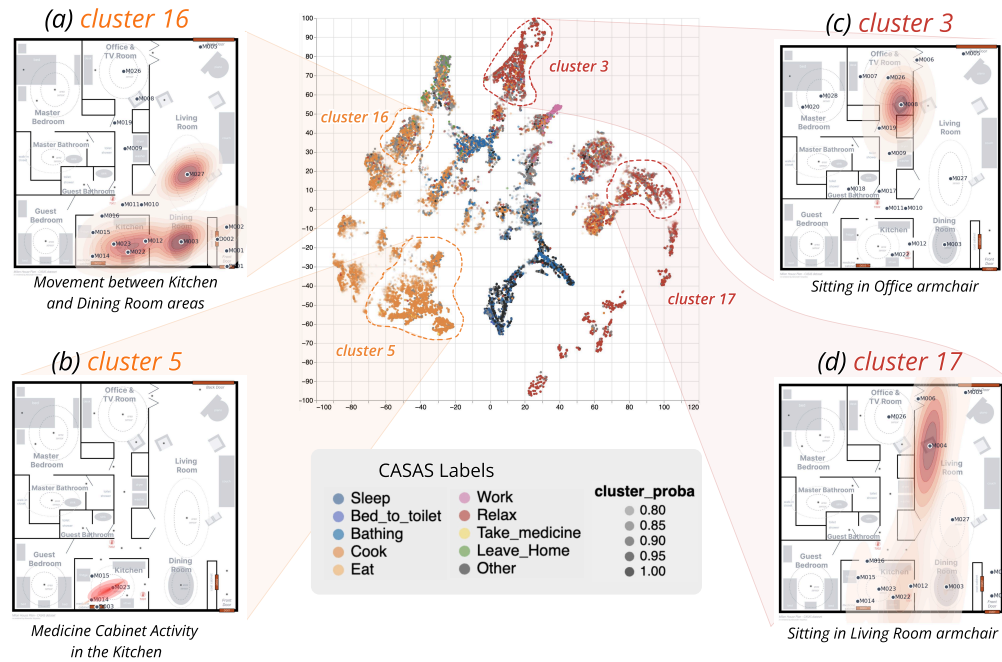


Figure 1: VISAR enables interpretable activity recognition by replaying sensor activations on 2D floor plans. Researchers used VISAR to analyze clusters from a black-box model in the Milan home (CASAS dataset). Although clusters 5 and 16 both had “Cook” label, VISAR revealed distinct patterns: (a) walking between kitchen and dining room vs. (b) using medicine cabinet. Similarly, within the broad “Relax” label, VISAR helped distinguish (c) sitting in TV chair from (d) sitting in living room corner. VISAR let experts relabel nuanced, house-specific activities by grounding sensor traces in real space and time.

Abstract

The effective deployment of human activity recognition (HAR) models in smart homes is hampered by the lack of labeled data and challenges in model interpretation. We introduce VISAR, a visualization and debugging tool designed to interpret and validate sensor-based HAR outputs. VISAR overlays sensor activation sequences on spatial maps with temporal replay, facilitating clear interpretations by domain experts. Two case studies, unsupervised activity discovery and longitudinal anomaly detection, show how VISAR supports debugging and informed human oversight. By enabling expert oversight of state-of-the-art HAR models, VISAR

fosters transparency, trust, and robustness, crucial for real-world deployment.

CCS Concepts

• **Human-centered computing** → **Visualization systems and tools**; *Ubiquitous and mobile computing design and evaluation methods*; *Interactive systems and tools*.

Keywords

Activity Recognition, Smart Homes, Explainable AI, Visualization

ACM Reference Format:

Alexander Karpekov, Sonia Chernova, and Thomas Plötz. 2025. VISAR: Visualization and Interpretation of Sensor-based Activity Recognition for Smart Homes. In *Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp Companion '25)*, October 12–16, 2025, Espoo, Finland. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3714394.3756319>



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ACM ISBN 979-8-4007-1477-1/2025/10
<https://doi.org/10.1145/3714394.3756319>

1 Introduction

Human Activity Recognition (HAR) from ambient smart home sensors promises early warning of falls, cognitive decline, and loss of independence [2, 4, 13, 17, 18]. State-of-the-art pipelines, from supervised CNN / LSTM classifiers to transfer learning systems such as TDOST [20], are trained and benchmarked on richly annotated datasets (e.g., CASAS [9], Marble [3]). These labels make offline evaluation easy but can become problematic when it comes to real home deployment.

In practice, every house differs: floor plans, sensor placements, and daily routines vary widely, and no ground truth labels exist to verify predictions. An episode from the OHSU DETECT-AD study [16] illustrates this issue: a sensor tagged “laundry room” fired repeatedly at night, and an ML pipeline labeled the surge as late night laundry. Spatial–temporal replay with our VISAR tool showed the sensor actually sat in a pass through hallway next to a guest room. The activity reflected a visiting relative, not a laundry habit. Such misclassifications could misdirect caregivers.

Prior visual tools (e.g., *SituVis* timelines or *Visual Process Maps*) aid general sensor analytics but omit live floor plan context and fine-grained interaction. We therefore present VISAR, a model-agnostic framework that overlays raw activations and model outputs on a 2-D house plan, letting clinicians and ML engineers replay, inspect, and label events in space and time.

We demonstrate VISAR’s versatility through two case studies: (i) **DISCOVER** [12]: VISAR let experts hand-label 20 unsupervised clusters in under one hour, revealing fine-grained activities such as “medicine-cabinet reach” hidden inside generic *Kitchen Activity* labels; and (ii) **Anomaly Detection Debugging**: When a longitudinal anomaly occurred, VISAR helped to pinpoint the sensors and rooms that changed and identify whether the change was associated with a true health-related shift.

Our **contributions** are as follows:

- *Generalizable visual framework*: VISAR supports supervised, unsupervised, transfer-learning, and anomaly-detection models.
- *Interactive, granular debugging*: Experts can trace predictions to specific sensor events and contexts.
- *Real-world validation*: Field studies show how VISAR prevents critical labeling misinterpretations.

2 Related Work

Research on smart-home activity recognition spans (i) fully supervised and transfer-learning pipelines that require labeled data, and (ii) a growing body of Explainable AI (XAI) tools that seek to make those pipelines transparent to end-users.

Supervised and Transfer-Learning Human Activity Recognition. Early ambient-sensor Human Activity Recognition (HAR) relied on classic machine-learning techniques, including Random Forests [19], Hidden Markov Models (HMMs) and Support Vector Machines (SVMs) [10]. Modern deep systems replace hand-crafted features with Deep Neural Networks [1] and Long Short-Term Memory networks (LSTMs) [15].

To reduce the need to obtain per-home ground truth labels, transfer-learning frameworks (e.g., TDOST [20]) convert each sensor event into a natural-language “prompt” so that a single pre-trained language model can be fine-tuned and reused across multiple houses. These approaches achieve state-of-the-art F_1 when trained and tested on publicly available labeled datasets such as CASAS [9]. However, they assume the target environment’s sensors map cleanly onto the source ontology. This assumption oftentimes breaks down in heterogeneous, unlabeled real-world deployments.

Explainable-AI for Smart-Home Models. Most XAI work for ambient sensors adapts generic tabular explainers: *LIME* perturbs inputs locally to rank influential sensors, while *SHAP* computes game-theoretic attributions (e.g., applied to a CASAS LSTM in [11]). Though helpful for developers, these textual feature lists ignore room layout, leaving researchers uncertain how an “important” activity sensor maps to real behavior. Several domain-specific visual tools attempt to close that gap:

- *Situvis* [8] overlays raw event timelines but treats space abstractly;
- *CASASviz* [6] renders sensor heat-maps on a floor plan yet lacks live replay or model overlays;
- *Visual Process Maps* [14] mine routine graphs but detach sequences from geography;
- Baudisch *et al.*’s anomaly dashboard [5] outputs rule-based textual rationales without spatial context.

No existing tool combines full floor plan animation with model predictions, nor do they support debugging heterogeneous outputs from clustering, supervision, and longitudinal change detection within a single interface.

VISAR targets this gap directly. By animating raw sensor firings and coloring them with any model’s labels, clusters, or anomaly scores, the tool lets ML engineers and clinicians inspect individual sensor activation sequences. Unlike prior XAI methods, VISAR offers spatial–temporal replay for an activity sample of any length or granularity, providing a practical bridge between black-box smart-home ML models and domain experts.

3 VISAR Visualization Framework

VISAR is a general, easy-to-use tool for explaining and debugging smart home HAR models. Unlike *Situvis* [8] or *Visual Process Maps* [14], it fuses spatial *and* temporal views so that practitioners can interpret predictions quickly.

Spatial–Temporal View. Sensor events replay on the floor plan while a timeline scroller shows order and duration, making it simple to confirm a “cooking” label or spot sensors firing in the wrong room.

Interactive Debugging. Clicking any event reveals raw readings and baseline comparisons, exposing misinterpretations that arise when models trained on one dataset (e.g., CASAS [9]) meet different layouts (e.g., OHSU Detect homes [16]).

Clinically Focused. A caregiver-oriented UI turns opaque ML outputs into clear, actionable insights for resident care.

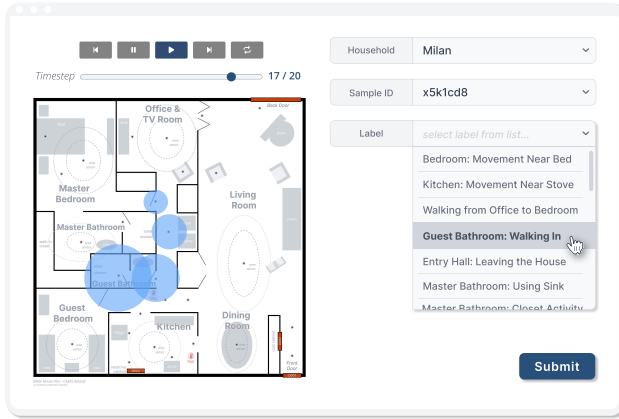


Figure 2: VISAR annotation screen used to label cluster centroids from an unsupervised learning study [12]. Panel on the left allows to replay individual sensor activation sequences on a 2-D floor plan (motion sensors flash in the hallway and bathroom), while a drop-down panel on the right lets the expert select or type a fine-grained activity label.

4 Case Studies

Within the AI-Caring Institute’s smart-home program [7], VISAR has already been integrated into two active pipelines: (i) the label-free DISCOVER clustering workflow [12], where VISAR lets experts assign nuanced, home-specific labels to cluster centroids, and (ii) a longitudinal Change-Detection Score (CDS) monitor, in which VISAR visually contrasts baseline and outlier days so experts can decide whether each spike reflects a clinically meaningful change or a benign context shift.

4.1 Case Study 1: DISCOVER - Unsupervised Activity Recognition

To illustrate VISAR’s practical value, we applied it to the *DISCOVER* pipeline, which mines fine-grained sub-activities from unlabeled smart-home data via unsupervised clustering and minimal human-in-the-loop annotation [12].

Background and Motivation. Unsupervised methods can uncover nuanced human behaviors without the heavy cost of manual labels, yet their cluster outputs are abstract and hard to interpret. Traditional visualizations rarely supply the spatial-temporal detail needed for meaningful annotation. In *DISCOVER* study [12], we trained a self-supervised model that created rich embeddings for sensor activation sequences and fine-tuned that model to create 20 distinct clusters. Since the approach assumed no access to ground truth labels, we identified representative cluster centroid sequences and turned to human labelers to annotate them.

Using VISAR for Cluster Interpretation. To evaluate interpretability and label consistency, we conducted a structured annotation study using VISAR. For each of the 20 clusters produced by the clustering pipeline, we randomly sampled 5 representative sensor-activation sequences (centroid-nearest examples), resulting in 100 samples per home. Each sample was independently labeled by two annotators

(200 ratings total per dataset). Annotators were graduate students with no prior experience in activity recognition and were provided with a hierarchical taxonomy of activity labels. They reviewed each sample using VISAR interface (Figure 2) and assigned the most appropriate label based on movement patterns and location context. We used inter-rater agreement using Cohen’s Kappa and cluster-level consistency using Fleiss’ Kappa to confirm the robustness of interpretation across different levels of abstraction. See full detail in [12].

With VISAR, annotators achieved very high inter-rater agreement on fine-grained labels, easily distinguishing subtle behavioral differences such as “movement near stove” versus “movement near medicine cabinet” or separate relaxation spots (living-room armchair vs. TV room). The immediate spatial-temporal context supplied by VISAR, which was missing in earlier supervised HAR methods, proved essential for clear, consistent interpretation.

4.2 Case Study 2: Longitudinal Behavior Anomaly Detection and Debugging.



Figure 3: VISAR helped debug an anomaly flagged in the Detect dataset [16]. Using VISAR, research team identified that simultaneous activations in the master bedroom and living-room couch, alongside a nearby “laundry room” sensor, signaled a guest presence, and not a clinically relevant shift in resident behavior.

We next evaluated VISAR in a longitudinal setting using a data-driven *Change Detection Score* (CDS), which tracks sensor activation trends over months to flag anomalies and significant behavioral shifts, which is key for spotting early cognitive or physical decline in older adults.

Background and Motivation. Interpreting longitudinal anomalies is difficult: a deviation might signal genuine clinical change or merely reflect contextual noise. Existing visual tools rarely supply the spatial–temporal detail required to resolve such ambiguities.

Debugging Anomalies with VISAR. After ML pipeline flagged a possible anomaly, VISAR helped clinicians separate *genuine behavioral changes* (e.g., a true rise in night-time kitchen visits) from *context-dependent artifacts*. In one Detect [16] home the ML model labeled a surge of nocturnal events as “laundry room activity.” VISAR’s replay (Figure 3) showed that the laundry motion activity sensor was mounted beside the living room couch and was activated whenever somebody watched late-night TV, while the primary resident remained in bed. By visualizing simultaneous bedroom (including the in-bed EMFIT sensors), hallway, and living-room activations, VISAR revealed night-time wandering and an extra occupant—not midnight laundering—so the spike was dismissed as a non-clinical change.

VISAR gave clinicians the fine-grained context needed to separate genuine behavioral change from sensor-placement artifacts, avoiding misinterpretations and inappropriate interventions. This case underscores VISAR’s value for real-world caregiving and clinical decision-making.

5 Discussion and Future Work

We presented VISAR, a generalizable visual-analytics tool for explaining and debugging smart-home activity-recognition models. VISAR sets itself apart from prior work in three key respects:

- **Generalisable visual framework.** VISAR accommodates supervised, unsupervised, transfer-learning, and anomaly-detection pipelines, rather than targeting a single workflow or dataset.
- **Interactive, granular debugging.** Domain experts can replay sensor sequences, drill into individual activations, and pinpoint the exact source of errors or misinterpretations.
- **Clinically relevant interpretation.** The interface is tailored for caregivers and clinicians, ensuring that AI outputs are conveyed in an intuitive, actionable form.

We validated VISAR in two real-world scenarios: unsupervised sub-activity discovery (*DISCOVER*) and longitudinal monitoring and anomaly detection, demonstrating that it can surface critical misinterpretations overlooked by advanced transfer-learning models. By supplying the spatial–temporal context missing in earlier tools, VISAR enables accurate understanding of smart home data even when ground truth labels are absent and spatial layouts vary widely.

As smart home sensing becomes further embedded in healthcare, bridging the gap between powerful yet opaque ML models and the transparency required in clinical practice will be essential. Tools like VISAR can provide that bridge.

Acknowledgments

This work was partially supported by NSF Grant IIS-2112633.

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